

Data Request

STADIOEquities — Account Activation Prediction Project

Module: CAP182 — Capstone (School of Information Technology)

Assignment: SS1 — Part C (Data Request)

Client: STADIOEquities (Retail Investing & Fintech)

Introduction

This document specifies the data required from STADIOEquities to develop a machine learning model that predicts whether a newly registered account will make its first deposit within a 30-day activation window. The data request is organized into five tables corresponding to the key data sources identified in the STADIOEquities information pack. Each column includes the column name, data type, example value, and additional information to guide data extraction.

Table 1: Account & Registration Data

Core account-level data covering the registration journey from sign-up through KYC completion and first deposit. This is the primary data source for defining the target variable (activated vs. not activated within 30 days) and extracting the early behavioural features.

Column Name	Data Type	Example	Additional Information
Account ID	String, UUID	a3f8c1d2-4e5b-6c7d	Unique identifier for each registered account. Primary key for joining with other tables.
Registration date-time	Datetime (UTC)	2025-03-15 14:23:00	Exact timestamp when the account was created. Format: yyyy-MM-dd HH:mm:ss UTC.
Registration source	String, categorical	Web	Platform used for registration: Web, iOS App, Android App.
Acquisition channel	String, categorical	Google Ads	Marketing channel through which the user was acquired. Valid values: Google Ads, Facebook Ads, Organic Search, Referral, Direct, Influencer, PR, Other.
Campaign ID	String	CAMP-2025-Q1-042	Identifier of the specific marketing campaign. Nullable if organic acquisition.
Onboarding steps completed	Integer	5	Count of onboarding steps completed (out of total steps in current flow). Range: 0 to current max steps.
KYC status	String, categorical	Completed	Current KYC status: Not Started, In Progress, Completed, Rejected.
KYC completion date-time	Datetime (UTC)	2025-03-16 09:12:00	Timestamp when KYC was fully completed. Nullable if not completed.
KYC time-to-complete (hours)	Numeric, float	18.75	Hours between registration and KYC completion. Nullable if KYC not completed.
First login date-time	Datetime (UTC)	2025-03-15 14:30:00	Timestamp of first login after registration. Should be shortly after registration.
First deposit date-time	Datetime (UTC)	2025-03-18 11:45:00	Timestamp of first deposit. Nullable if no deposit made. Critical for defining the target variable.

Column Name	Data Type	Example	Additional Information
First deposit amount (ZAR)	Numeric, float	500.00	Amount of first deposit in South African Rand. Nullable if no deposit.
Time-to-first-deposit (hours)	Numeric, float	69.25	Hours between registration and first deposit. Nullable if no deposit.
Activated (30-day flag)	Boolean	True	Target variable: True if first deposit within 30 days of registration, False otherwise.
Days since registration	Integer	45	Elapsed days since registration date. Used for cohort analysis.
Account status	String, categorical	Active	Current status: Active, Dormant, Closed, Suspended.

Table 2: App & Web Behavioural Data

Session-level and event-level data capturing user interactions with the STADIOEquities platform during the first 30 days after registration. These features capture the intensity and nature of early engagement, which is expected to be strongly predictive of activation.

Column Name	Data Type	Example	Additional Information
Event ID	String, UUID	evt-7a9b2c3d	Unique identifier for each behavioural event logged.
Account ID	String, UUID	a3f8c1d2-4e5b-6c7d	Foreign key linking event to the account.
Event date-time	Datetime (UTC)	2025-03-15 14:35:22	Timestamp of the event. Format: yyyy-MM-dd HH:mm:ss UTC.
Event type	String, categorical	PageView	Type of event. Valid values: PageView, ScreenView, Click, ButtonPress, Search, FormSubmit, VideoWatch, FeatureUse, Error, NotificationOpen.
Page / Screen name	String	BrowseStocks	Name of the page or screen where the event occurred.
Days since registration	Integer	0	Number of days between this event and the account's registration date.
Session ID	String	sess-f4e5d6c7	Unique identifier for the session. Events within the same session share this ID.
Session duration (seconds)	Numeric, integer	245	Duration of the session in seconds. Aggregated per session.
Number of sessions (first 7 days)	Integer	12	Total number of app/web sessions in the first 7 days after registration. Aggregated feature.
Number of sessions (days 8-30)	Integer	5	Total number of sessions between days 8 and 30 after registration.
Deepest onboarding screen reached	String	FundYourAccount	The furthest onboarding screen the user navigated to during the first session.
Tutorial completion	Boolean	True	Whether the user completed the in-app tutorial: True or False.
Stocks searched	Integer	3	Number of distinct stocks searched in the first 30 days.
Portfolio page views	Integer	7	Number of times the portfolio/watchlist page was viewed in the first 30 days.
Deposit page views	Integer	2	Number of times the deposit/funding page was viewed in the first 30 days.
Notification opens (first 7 days)	Integer	4	Number of push notifications or emails opened in the first 7 days.
Error events (first 7 days)	Integer	1	Number of error events encountered in the first 7 days.
Days since last session	Integer	2	Number of days since the user's most recent session (as of day 30).

Table 3: Client Demographics

Demographic and profile information collected at registration. These features capture the user's background, financial goals, and stated preferences, which may correlate with activation likelihood.

Column Name	Data Type	Example	Additional Information
Account ID	String, UUID	a3f8c1d2-4e5b-6c7d	Foreign key linking demographics to the account.
Age	Integer	29	Age in years at the time of registration. Range: 18 to 100.
Age group	String, categorical	25-34	Binned age group. Valid values: 18-24, 25-34, 35-44, 45-54, 55-64, 65+.
Gender	String, categorical	Male	Self-reported gender. Valid values: Male, Female, Non-binary, Prefer not to say, Not specified.
Province	String, categorical	GP	South African province. Valid values: GP, EC, FS, KZN, LP, MP, NC, NW, WC.
Country	String	South Africa	Country of residence for non-SA clients. If South Africa, matches province.
Employment status	String, categorical	Employed	Self-reported. Valid values: Employed, Self-employed, Student, Unemployed, Retired, Other.
Income bracket	String, categorical	R200k-R400k	Self-reported annual income. Valid values: Below R100k, R100k-R200k, R200k-R400k, R400k-R600k, Above R600k, Prefer not to say.
Investment experience	String, categorical	Beginner	Self-reported experience. Valid values: Beginner (0-1 years), Intermediate (1-5 years), Advanced (5+ years), Professional.
Investment goal	String, categorical	Long-term growth	Stated primary goal. Valid values: Long-term growth, Savings, Retirement, Short-term gains, Education fund, Other.
Risk appetite	String, categorical	Medium	Risk tolerance at sign-up. Valid values: Low, Medium-Low, Medium, Medium-High, High.
Referral code	String	NULL	Referral code used at sign-up, if any. Nullable.

Table 4: Marketing & Nudge Engagement Data

Data on marketing communications sent to the user and their engagement with each communication. This captures the touchpoints between STADIOEquities and the new registrant during the critical activation period.

Column Name	Data Type	Example	Additional Information
Communication ID	String, UUID	comm-b1c2d3e4	Unique identifier for each communication sent.
Account ID	String, UUID	a3f8c1d2-4e5b-6c7d	Foreign key linking communication to the account.
Communication date-time	Datetime (UTC)	2025-03-16 08:00:00	When the communication was sent. Format: yyyy-MM-dd HH:mm:ss UTC.
Communication channel	String, categorical	Email	Channel used. Valid values: Email, Push Notification, SMS, In-app Message.
Communication type	String, categorical	Onboarding	Type of communication. Valid values: Welcome, Onboarding, Nudge, Promotional, Educational, Product Update, Re-engagement.

Column Name	Data Type	Example	Additional Information
Days since registration	Integer	1	Number of days since registration when this communication was sent.
Opened	Boolean	True	Whether the communication was opened (email open, push notification tapped). Nullable for SMS.
Clicked	Boolean	True	Whether any link in the communication was clicked. Nullable if not opened.
Opened (within 1 hour)	Boolean	True	Whether the communication was opened within 1 hour of being sent.
Total emails sent (first 30 days)	Integer	8	Count of emails sent in first 30 days. Aggregated feature.
Total emails opened (first 30 days)	Integer	5	Count of emails opened in first 30 days.
Total push notifications sent	Integer	12	Count of push notifications sent in first 30 days.
Total push notifications opened	Integer	7	Count of push notifications opened in first 30 days.
Total nudges sent	Integer	5	Count of onboarding nudge messages sent in first 30 days.
Cost per acquisition (ZAR)	Numeric, float	180.00	Marketing cost attributed to acquiring this account in South African Rand.

Table 5: Product & Subscription Data

Data on premium subscription uptake and value-added product interactions during the first 30 days. While activation and premium uptake are sequential (deposit first, then subscribe), exposure to premium offers during onboarding may signal engagement level.

Column Name	Data Type	Example	Additional Information
Account ID	String, UUID	a3f8c1d2-4e5b-6c7d	Foreign key linking to the account.
Premium subscription status	String, categorical	Free	Current subscription tier. Valid values: Free, Premium Monthly, Premium Annual.
Premium offer shown	Boolean	False	Whether a premium subscription offer was presented during onboarding.
Premium offer click	Boolean	False	Whether the user clicked on a premium subscription offer.
Premium subscription date	Datetime (UTC)	NULL	Date when user subscribed to premium. Nullable if on free tier.
Value-added products viewed	Integer	1	Number of value-added product pages viewed in first 30 days (e.g., offshore accounts, tax-free accounts).
Product comparison page views	Integer	0	Number of times the product comparison page was visited.

Notes

- All datetime fields must be in UTC and formatted as yyyy-MM-dd HH:mm:ss.
- Null values should be represented as empty fields (not 'NULL', 'N/A', or 'null' as strings) for all tables.
- Tables 2, 4, and 5 contain event-level data that will need to be aggregated to the account level for modelling.
- The aggregated features (e.g., 'Number of sessions (first 7 days)') are included in the event-level tables for convenience but can also be derived from the raw event data.

- The 'Activated (30-day flag)' in Table 1 is the target variable. For model training, we request this field for historical data. For production inference, it will be derived.
- We request data covering at least the past 24 months of registrations to ensure sufficient sample size and seasonal coverage.
- Any personally identifiable information (PII) such as names, email addresses, and phone numbers should be excluded or anonymized.